

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. CHB-608: Atomic and Molecular Structure and Spectroscopic Techniques

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Sketch the symmetry elements present in: (i) H₂O and (ii) NH₃.
- (b) Write the radial, angular wave equations for the Hydrogen atom.
- (c) What are the basic principles of the Valence Bond Theory?
- (d) What is Fermi resonance in IR spectra?
- (e) What are metastable ions in mass spectrometry?

Question 2.

- (a) Explain the molecular orbital treatment of the H₂ molecule. Deduce the expressions for bonding and antibonding molecular orbitals. [6]
- (b) Derive Schrödinger's wave equation for a particle in a 3-D box and calculate the degeneracy of energy levels. [8]

Question 3.

- (a) How will you distinguish the following pairs with the help of IR spectroscopy? [6]
 - (i) Intermolecular vs. intramolecular hydrogen bonding (dilution effect)
 - (ii) o-Hydroxybenzoic acid and p-hydroxybenzoic acid
- (b) Discuss the basic principles of the origin of the ¹H-NMR spectrum. Why is TMS used as a reference? [8]

Question 4.

- (a) Define: chromophore, auxochrome, bathochromic shift, and hypsochromic shift. Give examples of each. [6]
- (b) Predict the masses of the ions produced in the mass spectrum of 2-hexene by β-fragmentation. [4]

- (c) What are the masses of the ions produced in the mass spectra of 4-*n*-butyltoluene by McLafferty rearrangement and benzylic fission? [4]